

DISCRETE-TIME FILTER DESIGN

SAMPLE PROBLEMS

ENSC 429 – Digital Signal Processing

School of Engineering Science
Simon Fraser University
Burnaby, BC, Canada



DT IIR FILTER DESIGN

Design a minimum-order Butterworth DT filter to meet the following specifications:

$$\begin{aligned} 0.9 \leq |H(e^{j\omega})| \leq 1 & \quad 0 \leq |\omega| \leq 0.1\pi \\ |H(e^{j\omega})| \leq 0.1 & \quad 0.5\pi \leq |\omega| \leq \pi \end{aligned}$$

- a) Design the DT filter using the impulse invariance method applied to the corresponding CT Butterworth filter
 - b) Show a direct form I realization of the DT filter
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- Solution: [see notes](#)

DT FIR FILTER DESIGN

DT FIR filter specifications are given as follows:

$$\begin{aligned} 0.98 \leq H(e^{j\omega}) \leq 1.02 & \quad 0 \leq |\omega| \leq 0.63\pi \\ -0.15 \leq H(e^{j\omega}) \leq 0.15 & \quad 0.65\pi \leq |\omega| \leq \pi \end{aligned}$$

- a) Which of the windows we have covered in class (rectangular, Bartlett, Hann, Hamming, Blackman, Kaiser) can be used to satisfy the specifications? Why?
 - b) Find the parameters of the Kaiser window that would meet the specifications
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- Solution: [see notes](#)

DT FIR FILTER DESIGN BY WINDOWING

TABLE 7.2 COMPARISON OF COMMONLY USED WINDOWS

Type of Window	Peak Side-Lobe Amplitude (Relative)	Approximate Width of Main Lobe	Peak Approximation Error, $20 \log_{10} \delta$ (dB)	Equivalent Kaiser Window, β	Transition Width of Equivalent Kaiser Window
Rectangular	-13	$4\pi/(M+1)$	-21	0	$1.81\pi/M$
Bartlett	-25	$8\pi/M$	-25	1.33	$2.37\pi/M$
Hann	-31	$8\pi/M$	-44	3.86	$5.01\pi/M$
Hamming	-41	$8\pi/M$	-53	4.86	$6.27\pi/M$
Blackman	-57	$12\pi/M$	-74	7.04	$9.19\pi/M$